

30 April 2026

Dr. Siti Nur Khadijah Aishah Binti Ibrahim

Lecturer, SCSE1203 – Software Engineering Principles
Malaysia-Japan International Institute of Technology (MJIIT)
Universiti Teknologi Malaysia, Kuala Lumpur

Dear Dr. Aishah,

The following is our proposal for the Virtual Internship Portal (VIP for short),

This document is a continuation from our Project Charter. We developed this proposal through user surveys conducted with twenty students from different year groups, informal discussions with two local companies willing to act as proxy stakeholders (More on that later), and a low fidelity prototype built using Figma to test our assumptions before settling on any particular design direction. We then compared three feasible technical approaches we could potentially take against one another before putting forward a recommendation we believe is actually supported by the evidence, not simply the most appealing option on paper.

In this proposal we aim to be transparent about what we do not know, where the data is limited, and where our decisions are subjective. The survey sample is quite small, company feedback came from just 2 companies rather than a wider industry base, and our prototype is mainly just designs and not an actual working system. We feel this should be flagged upfront rather than presenting findings as more conclusive than they actually are.

We are happy to elaborate on any sections that require more justification, or where our reasoning is unclear, during our next (online D:) sprint review or during our presentation.

Thank you for your continued guidance throughout this project.

Yours sincerely,

Group 1 – BOLEH

Nahid Hasan Rayan (Product Owner)
Jobayer Alam (Scrum Master)
Khalid Waleed Issa (Development Team)
Zay Yar Shin (Development Team)
Ghazi Anwar Bin Ahmad Razif (Development Team)

Malaysia-Japan International Institute of Technology (MJIIT)

Universiti Teknologi Malaysia, Kuala Lumpur

SCSE1203 – Software Engineering Principles

SYSTEMS PROPOSAL

Virtual Internship Portal

No.	Full Name	Matric No.	Scrum Role	Weekly Hours
1	Nahid Hasan Rayan	A25MJ0186	Product Owner	8–10 hrs
2	Jobayer Alam	A25MJ0184	Scrum Master	3–5 hrs
3	Khalid Waleed Issa	A25MJ0185	Dev Team	5–8 hrs
4	Zay Yar Shin	A25MJ0187	Dev Team	5–8 hrs
5	Ghazi Anwar Bin Ahmad Razif	A25MJ5072	Dev Team	5–8 hrs

Group Name: Group 1 – BOLEH

Course: SCSE1203 – Software Engineering Principles

Faculty: Malaysia-Japan International Institute of Technology (MJIIT)

Submission Date: 30 April 2026

Lecturer / Evaluator: Dr. Siti Nur Khadijah Aishah Binti Ibrahim

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4. Executive Summary

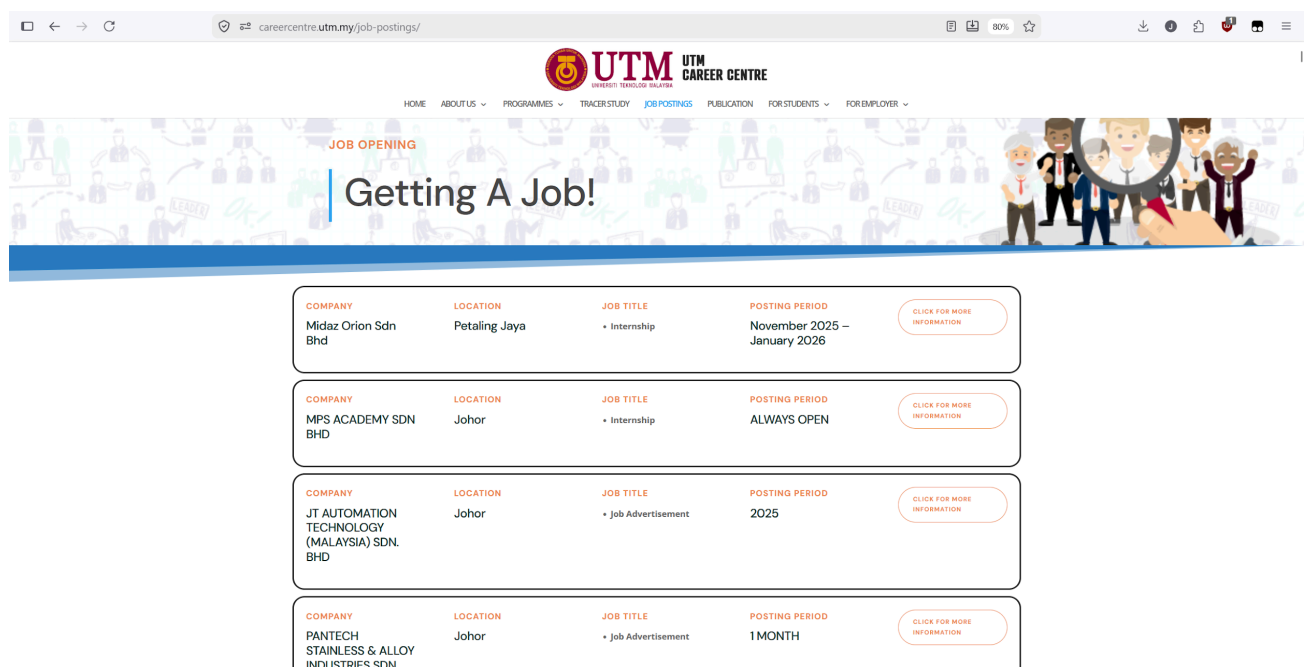
4.1 What we wanted to do

Choosing Virtual Internship Portal as a project was not something we wanted to do just because it was easy. The aspect of future internships is something that is genuinely scary.

- Where would we get our placements?
- What is the work culture there?
- Would we have a future there as potentially new hires?
- What skills should we focus on growing now and what things are just noise?
- Are these positions paid or unpaid?

The list of questions we have on our mind just goes on.

We, currently in the middle of our second semester, are still unsure about the most critical aspect of our bachelors: the first, real work experience most of us will get, which will dictate our future job offers and career path. We are sure UTM will oversee this matter perfectly as they have done so for our seniors and those who came before us. However, the point we are trying to make is the lack of transparency in something so important. Although UTM does have a career centre website.



Unfortunately, it is outdated. It is slow to load, resulting in a bad user experience. There is no way to filter by location, company type, job posting, work period etc. These are options you would expect from a modern website which are not present. We were also sad to see some of the pages were completely blank.

We wanted to make sure we fully understood the problem before proposing any solution. This proposal sets out what we found through a small survey and stakeholder engagement, which three technical options we genuinely explored, and the reasoning behind our eventual recommendation. Rather than simply listing conclusions, we have tried to show our working. Some of the decisions that might seem odd will make more sense with context behind them.

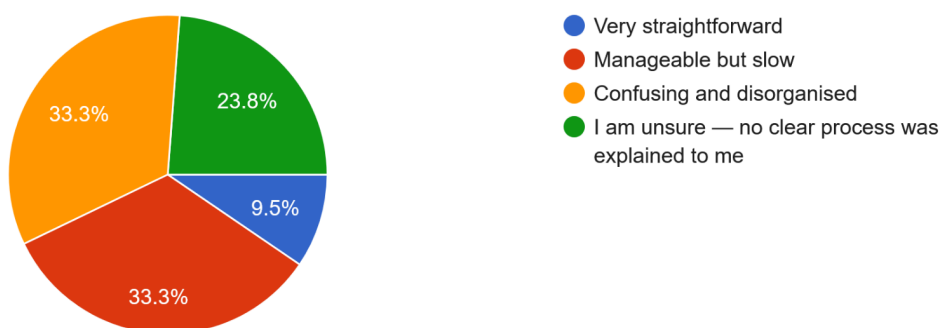
4.2 What We Found

In total we contacted 30 students/alumni. Of those 30, twenty one completed our structured Google Forms survey in full. The other nine preferred to engage through informal discussion rather than through the form. We also carried out short semi structured interviews with representatives from two local companies that agreed to speak with us as proxy stakeholders. **Damai Luxe Management** (Reg: 202503080173 (003711860-T)), a property management firm, and **Narma Transport Solutions** (Reg: JR0151935-H), a car rental agency.

The survey confirms what we suspected. The leading source for internship placements was external job portals such as JobStreet and LinkedIn (57.1%), while only 47.6% went through the university placement office. Meaning more than half of students are navigating the process entirely on their own. Over two thirds of respondents (66.6%) described the faculty process as confusing, disorganised, or simply never explained to them. The application volume data reinforces this, 66.7% applied to eight or more companies before securing a placement. Once applications were sent, 76.2% reported difficulty keeping track of them across different companies.

4. How would you describe the current internship application process at your faculty?

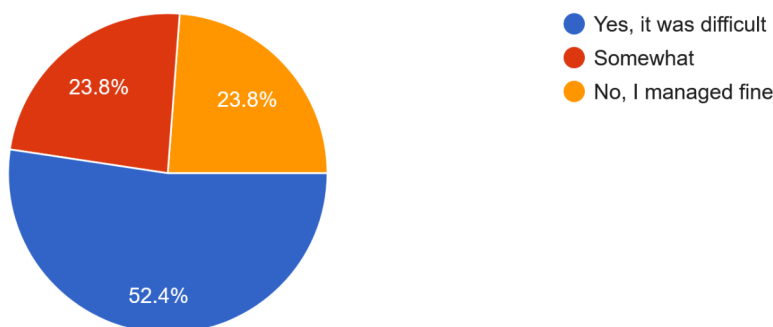
21 responses



Pictured: Over two thirds of respondents (66.6%) described the faculty process as confusing, disorganised, or simply never explained to them.

6. Did you face difficulty keeping track of your applications across different companies?

21 responses



Pictured: Only 23.8% of respondents found keeping track of their application manageable.

Both company contacts were candid. Damai Luxe Management and Narma Transport Solutions both said they have no shortage of applicants; the challenge is reaching UTM students specifically rather than drawing from a wide mix of institutions. Both said they would use a dedicated university portal if posting was free and easy. One put it plainly: "We would use it if we could trust the applicants are real students." That single comment shaped how we designed the admin-approval flow for company accounts.

4.3 What We Recommend

We looked at three options: a WordPress plugin approach, a no-code platform (Bubble.io), and a custom full-stack web application. The team recommends Option C: a custom application built with React on the frontend, Node.js on the backend, and PostgreSQL for data storage. Sections 7 and 8 detail the reasoning; in brief, it is the only option that fully covers what users asked for, fits within free-tier hosting, and aligns with what this course expects as a learning outcome.

The core system centres on five features:

Feature	Description
Student registration & profile	Sign-up, login, resume upload, skills list
Internship search & filter	Keyword, duration, and location filters
Company posting with admin approval	Structured listing workflow; companies verified before going live
Live application tracking	Status visible to student: pending, shortlisted, hired/rejected
Market-awareness panel	Ranked view of most-requested skills from active postings

A peer discussion forum is in-scope but lower priority. Full reasoning in Sections 7 and 8.

5. Outline of Systems Study with Appropriate Documentation

Seven steps were followed. The table below summarises each one.

#	Phase	Activity	Led By	Output
1	Problem ID	Informal discussions with 30 students/alumni and 2 company contacts.	Khalid, Jobayer	Problem statement; stakeholder list
2	Competitor Review	Reviewed LinkedIn, Internshala, UTM Career Portal. Gaps: no campus filter, no tracking, no skills data.	Nahid	Gap analysis (App. A)
3	User Survey	Google Forms to 30 students; 21 completed fully.	Ghazi, Shin	Survey results (App. B)
4	User Story Mapping	18 user stories derived and MoSCoW-prioritised.	Nahid	Prioritised backlog (App. C)
5	Prototyping	10 Figma screens designs created	Jobayer	Figma prototype (App. D)
6	Technical Feasibility	Verified free-tier viability: Vercel, Railway, Supabase.	Khalid, Nahid	Feasibility notes (Sec. 7)
7	Risk Assessment	Identified scope creep, uneven workload, limited company access.	All	Risk register (App. G)

5.1 Notes on Methodology

21 of 30 contacted completed the Google Form; the rest contributed informally. Respondents were not randomly selected most were people we know, which creates a mild bias toward wanting a solution. Company input came from two personal contacts only. All findings are treated as directional, not definitive.

6. Detailed Results of the Systems Study

6.1 Survey Highlights

Finding	Students (n=20)	Companies (n=2)
Search method	57.1% used external job portals such as JobStreet and LinkedIn; only 47.6% went through the university placement office	post on general platforms, struggle to reach UTM students specifically
Application volume	66.7% applied to 8 or more companies before securing a placement; 38.1% applied to more than 12	—
Process clarity	66.6% described the faculty internship process as confusing, disorganised, or never clearly explained to them	—
Tracking difficulty	76.2% faced difficulty keeping track of applications across companies (52.4% said it was outright difficult)	—
Top feature requested	90.5% want to submit and track applications in one place; 76.2% want real-time status updates	Both want admin verified student profiles
Portal adoption intent	100% want or would appreciate mobile access; 57.1% say it is essential	Both would use it if free and easy to post on

6.2 User Stories (Top Five by Priority)

ID	User Story	Priority	Sprint
US-01	"I want to register and log in securely as this website will contain very sensitive data about me"	Must Have	Sprint 1
US-02	"Having a place where i can upload my resume, list my skills etc would be nice"	Must Have	Sprint 1
US-03	"Having our company vetted out by an admin and giving us a verified option would encourage students to apply to us"	Must Have	Sprint 1
US-04	"Searching internships by location, skills required and timeline would be amazing to sort"	Must Have	Sprint 2
US-05	"As a company, I want to post a listing with all relevant details and a closing date."	Must Have	Sprint 2

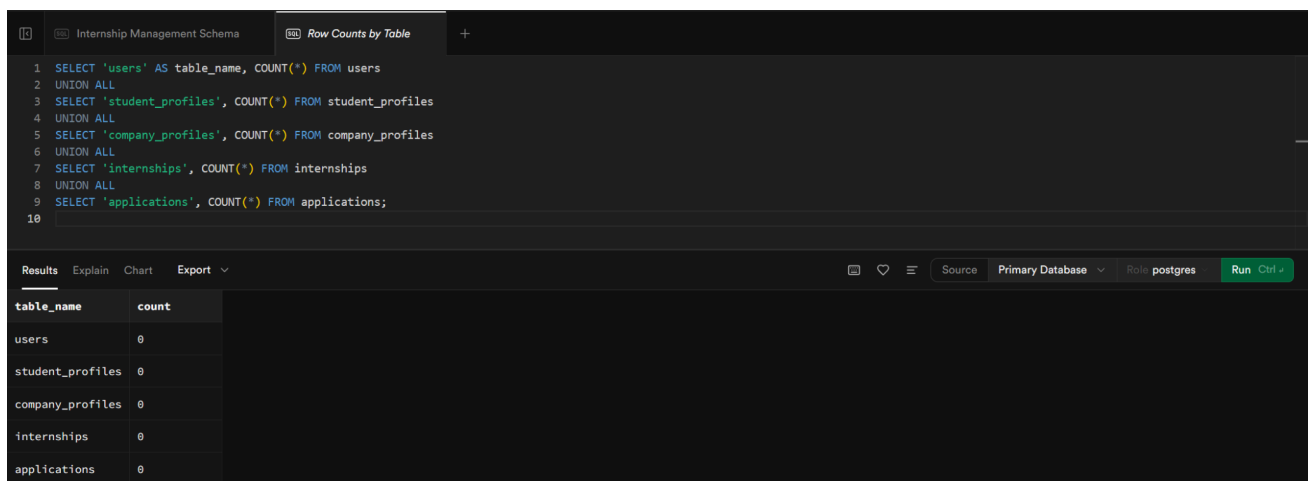
6.3 Database Schema

The database is built on PostgreSQL and structured around five core tables. The design keeps all roles in one users table, with separate profile tables branching out for students and companies.

PostgreSQL. Five tables:

Table	Key Fields	Notes
users	id, email, hashed_password, role, name, created_at	Single table for all roles (student / company / admin)
student_profiles	user_id (FK), matric_no, cgpa, resume_url, skills	One-to-one with users; skills as text array
company_profiles	user_id (FK), company_name, description, verified_by_admin	Cannot post until verified_by_admin = true
internships	id, company_id (FK), title, description, required_skills, duration, deadline, status	Status: draft active closed
applications	id, student_id (FK), internship_id (FK), status, applied_at, updated_at	Status: pending shortlisted rejected hired

All roles share one users table to keep authentication simple (single login flow). Full SQL in Appendix F.



```
1 SELECT 'users' AS table_name, COUNT(*) FROM users
2 UNION ALL
3 SELECT 'student_profiles', COUNT(*) FROM student_profiles
4 UNION ALL
5 SELECT 'company_profiles', COUNT(*) FROM company_profiles
6 UNION ALL
7 SELECT 'internships', COUNT(*) FROM internships
8 UNION ALL
9 SELECT 'applications', COUNT(*) FROM applications;
```

table_name	count
users	0
student_profiles	0
company_profiles	0
internships	0
applications	0

Picture: Screenshot showing the listed above tables in Supabase. Full SQL in linked Github repo in Appendix F.

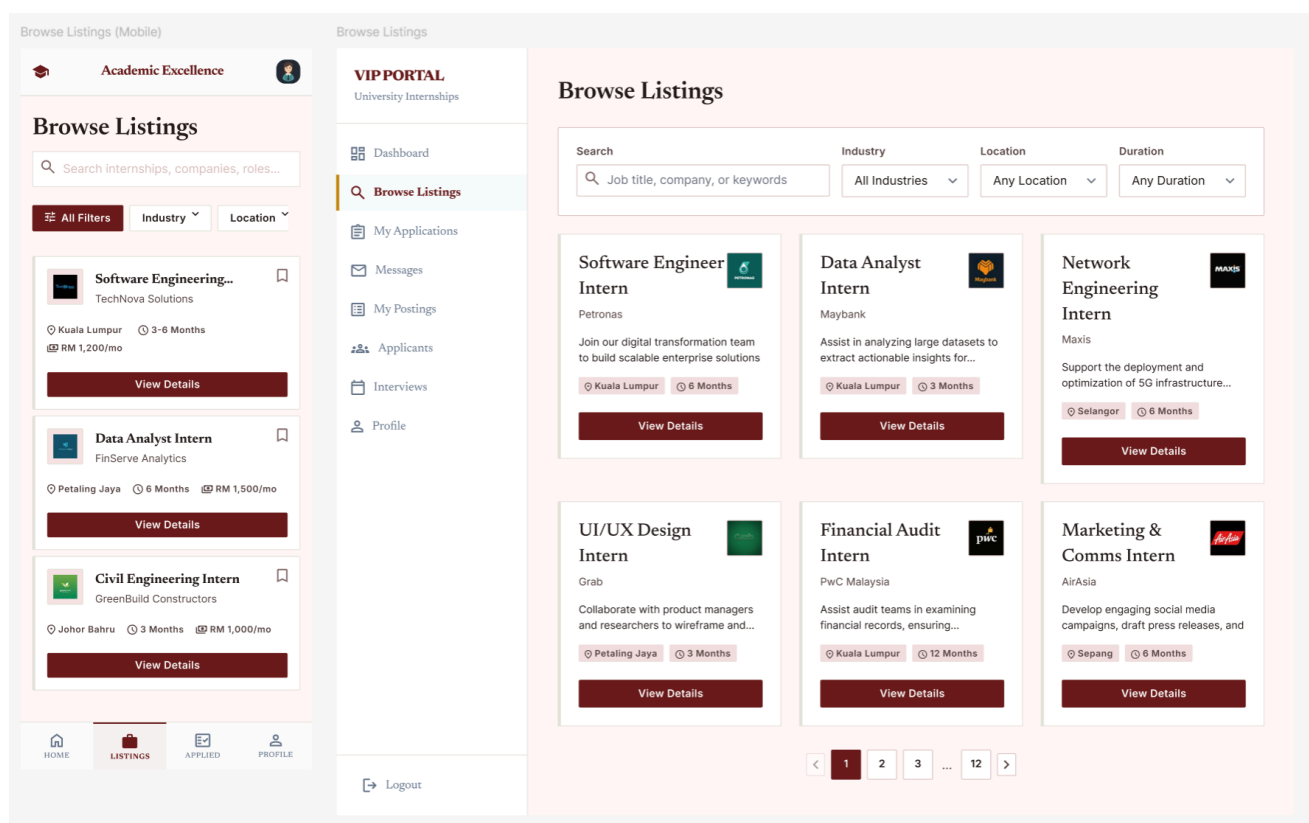
6.4 Low-Fidelity Prototype

Jobayer built 10 screens in Figma. While the original target was 5 screens, due to popular demand as according to the survey 57.1% of respondents said that they would "Definitely want it to be accessible on mobiles" while 42.9% of respondents said that "It would be nice but not essential" None of the respondents said that the desktop only view is fine. 5 extra mobile screens had to be made.

#	Screen	description
1	Browse Listings	search and filter internships by industry, location, and duration
2	Internship Details	full job description, responsibilities, requirements, and Apply button
3	Application status summary	application status summary, recent applications, and interview count
4	Application Form	CV upload, cover letter input, availability date, and submission
5	Supervisor Dashboard	applicant roster, pending reviews, and interview scheduling

Prototype link is provided in Appendix D. Below are some of the screenshots from Figma.

1 . Browse Listings



2 . Application Form

Application Form (Mobile)

Internship Portal

Step 2 of 3

Application Details

Application Form

Curriculum Vitae (CV)

Drag and drop your CV here, or click to browse.
PDF or DOCX, max 5MB

Cover Letter

Dear Hiring Manager...

Availability Start Date

mm/dd/yyyy

Why this company?

Briefly state your motivation...

Your application will be reviewed within 5-7 working days.

Back

Continue →

Application Form

VIP PORTAL

University Internships

Dashboard

Browse Listings

My Applications

Messages

My Postings

Applicants

Interviews

Profile

Logout

Application Form

STEP 2 OF 3

Please provide your supporting documents and availability for this position.

Curriculum Vitae (CV)

Upload your most recent CV in PDF format. Max size 5MB.

Click to upload or drag and drop

PDF only (max. 5MB)

Cover Letter

Introduce yourself and explain why you are a strong candidate for this specific role.

Dear Hiring Manager...

Availability Start Date

mm/dd/yyyy

Why this company?

Briefly summarize your motivation

Your application will be reviewed within 5-7 working days.

Back

Continue →

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3 . Supervisor View

Supervisor Dashboard (Mobile)

Supervisor Portal

Overview

Here is the latest status of your placements.

TOTAL APPLICATIONS

148

SHORTLISTED

24

PENDING REVIEW

82

Recent Applicants

VIEW ALL →

Nahid Hassan Rayan

Universiti Malaysia

Oct 12, 2026

BSc Computer Science

View Profile

Schedule Interview

Zay Yar Shin

Universiti Teknologi Malaysia

Oct 11, 2026

BEng Software Engineering

View Profile

Schedule Interview

DASHBOARD

APPLICANTS

PLACEMENTS

SETTINGS

Supervisor Dashboard

VIP PORTAL

UNIVERSITY INTERNSHIPS

Dashboard

Browse Listings

My Applications

Messages

My Postings

Applicants

Interviews

Profile

Logout

← BACK TO MY POSTINGS

Software Engineering Intern - Summer 2027

Reviewing applicants for posting ID #REQ-8892

TOTAL APPLICATIONS

148

SHORTLISTED

24

PENDING REVIEW

82

Applicant Roster

Search name or university...

Filter

STUDENT NAME	UNIVERSITY	PROGRAMME	DATE APPLIED	ACTIONS
Nahid Hasan Rayan	Universiti Malaysia	BSc Computer Science	Oct 12, 2026	View Profile Schedule Interview
Zay Yar Shin	Universiti Sains Malaysia	BEng Software Engineering	Oct 11, 2026	View Profile Schedule Interview
Sara Rahman	Universiti Teknologi Malaysia	BSc Information Technology	Oct 10, 2026	View Profile Schedule Interview
Kevin Lim	Universiti Putra Malaysia	BSc Computer Science	Oct 09, 2026	View Profile Schedule Interview

Showing 1 to 4 of 148 entries

Prev123...Next

4 . Student Dashboard

Student Dashboard (Mobile)

Welcome back,
Jobayer Alam

Bachelor of Computer Science • Year 3

APPLICATIONS SENT

12

INTERVIEWS SCHEDULED

3

OFFERS RECEIVED

1

Recent Applications

View All

Software Engineering Intern

Petronas • Apr 12, 2026

REVIEWED

Data Analyst Intern

Microsoft • Apr 02, 2026

PENDING

Frontend Developer Intern

Google • Mar 25, 2026

ACCEPTED

IT Support Intern

DHL • Mar 15, 2026

PENDING

DASHBOARD

APPLICATIONS

EXPLORE

PROFILE

Student Dashboard

VIP PORTAL

University Internships

Dashboard

Browse Listings

My Applications

Messages

My Postings

Applicants

Interviews

Profile

Logout

Welcome back, Jobayer Alam

Bachelor of Computer Engineering • Year 3

New Application

APPLICATIONS SENT

12

INTERVIEWS SCHEDULED

3

OFFERS RECEIVED

1

Recent Applications

View All →

Company	Role	Date Applied	Status	Action
Petronas	Software Engineering Intern	Apr 12, 2026	REVIEWED	⋮
Microsoft	Data Analyst Intern	Apr 02, 2026	PENDING	⋮
Google	Frontend Developer Intern	Mar 25, 2026	ACCEPTED	⋮
DHL	IT Support Intern	Mar 15, 2026	PENDING	⋮

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7. Systems Alternatives (Three Possible Solutions)

Four constraints shaped our evaluation: 5-week timeline, team of five with intermediate skills, zero budget, and a course that expects a demonstrated engineering process.

7.1 Option A – WordPress Plugin (e.g. WP Job Manager)

Advantages	Disadvantages
Very quick to set up (live in 1–2 days).	Student profiles, CGPA tracking, and status flow need paid plugins or custom PHP.
Free hosting and plugins.	Role-based access (student / company / admin) not available out of the box.
Non-technical admins can manage it.	Configuring a CMS does not develop the software engineering skills this course requires.
Familiar to non-technical stakeholders.	Output is WordPress-tied; cannot be moved to a GitHub portfolio.

Verdict: Viable for a simple tool, but falls short of our requirements without workarounds that consume sprint time.

7.2 Option B – No Code Platform (Bubble.io)

Advantages	Disadvantages
Visual dev; no server setup needed.	Complex logic (conditional flows, DB joins, RBAC) is harder than it looks.
Built-in database and deployment.	Zero code ownership; nothing to show on GitHub.
Good for quick proof-of-concept.	Free tier limits workflows and users; paid plan needed for serious use.
No infrastructure overhead.	Does not demonstrate the engineering skills the course assesses.

Verdict: Useful for teams without technical backgrounds. For us: no GitHub output, no code ownership, poor course fit.

7.3 Option C – Custom Full Stack Web App (React + Node.js + PostgreSQL)

Advantages	Disadvantages
Full control over features, data model, security.	Most development time required; little margin for scope creep.
Complete code ownership; deployable portfolio piece.	More failure points (auth bugs, deployment, DB errors).
Directly exercises course targets: REST API, DB normalisation, JWT, Scrum.	Requires careful upfront planning (Gantt + PERT, Appendix E).
All infrastructure free (Vercel, Railway, PostgreSQL).	Backend delays directly block frontend progress.

Verdict: Hardest option, but the only one that fully meets user needs and course requirements. Risk is manageable with our task-ownership structure.

8. Systems Analyst's Recommendation

The team recommends Option C. The decision was unanimous. The table below summarises the reasoning.

Factor	Why Option C
Course alignment	Options A and B do not produce the engineering artefacts the rubric requires. Option C produces schema diagrams, API docs, test results, and burndown charts that map directly to the assessment criteria.
User needs	Only Option C supports role-specific views, live application tracking, and admin-moderated company registration without custom workarounds.
Timeline feasibility	PERT critical path = 22 days (Appendix E). Fits the 5 week window if the backend stays on schedule. Nahid owns all backend, frontend carries 7 days of slack.
Team value	Every member leaves with a deployed full stack app (JWT auth, file uploads, normalised DB, Scrum process) as a portfolio piece.

The main risk is backend delay. We have mitigated this by concentrating backend ownership with Nahid and scheduling deployment in Sprint 3 (not Sprint 4) to surface issues early. Sprint 0 is done. Sprint 1 starts next week.

9. Proposal Summary

Item	Detail
Product	Web based internship portal for UTM students, companies, and admins
Tech stack	React (frontend) · Node.js/Express (backend) · PostgreSQL (database)
Hosting	Vercel (frontend) · Railway or Supabase (backend/DB) — all free tier
Sprints	5 sprints, Sprint 0–4, ~1 week each. Sprint 0 complete.
Budget	Zero cost
Critical path	22 days (Appendix E). Backend owned by Nahid.
Prototype	10 Figma screens Link in Appendix D.

9.1 What We Are Building

Role	Capabilities
Students	Centralised search with filters, application tracking with live status, skills market awareness panel, peer forum
Companies	Structured listing workflow, admin-moderated registration, per-posting applicant view
Administrators	Company account approval; forum moderation; system oversight

9.2 Key Risks and Controls

Risk	Prob.	Impact	Mitigation	Owner
Backend delay	Medium	High	Nahid owns all backend; Firebase Auth as JWT contingency	Nahid
Scope creep	High	Medium	Additions only via sprint review with PO sign-off	Nahid (PO)
Uneven workload	Medium	Medium	Daily standups; SM redistributes if anyone is blocked	Jobayer (SM)
Limited stakeholder input	High	Low	Two real contacts confirmed; simulated data as fallback	Dev Team
Deployment issues	Low	High	Deploy in Sprint 3 to surface problems early	Ghazi, Shin
Docs falling behind	Medium	Medium	Written incrementally each sprint; Shin + Khalid co-own	Shin, Khalid

9.3 Next Steps

Sprint 1 begins with database schema finalisation and authentication. Sprint review presentations to Dr. Aishah are scheduled at the end of every sprint.

10. Appendices

Appendix A – Competitor Analysis

Platform	What It Does	Key Gap for UTM Students
LinkedIn	Professional networking and global job board	Not campus-focused; no per-listing status tracking; requires polished profile
Internshala	Dedicated internship platform with student profiles and tracking	Not used in Malaysia; listings are India-centric
UTM Career Portal	General graduate listings managed by UTM	Targets full-time graduates, not interns; no self-service posting; no application tracking

Appendix B – Survey Summary

Question	Responses (n=21)
Q1: What is your current year of study?	Majority in Year 1 (28.6%) and Year 3 (23.8%). Year 4: 19%. Year 2 and Postgraduate: 14.3% each.
Q2: Have you completed or are you currently undergoing an internship?	57.1% have completed or are currently undergoing one. 33.3% have not yet.
Q3: How did you find your internship placement?	Top source: job portals such as JobStreet and LinkedIn (57.1%). University placement office and personal contacts tied at 47.6%.
Q4: How would you describe the current process at your faculty?	66.6% said confusing and disorganised (33.3%) or had no clear process explained to them (23.8%). Only 9.5% said very straightforward.
Q5: How many places did you apply to?	66.7% applied to 8 or more companies. 38.1% applied to more than 12.
Q6: Did you face difficulty tracking applications?	76.2% said yes or somewhat. Only 23.8% managed fine.
Q7: How long did you wait for faculty confirmation?	61.9% waited more than a month or are still waiting. Only 4.8% heard back within a week.
Q8: Which features would you want in the portal?	Top: submit and track applications in one place (90.5%). Real-time status updates (76.2%). Browse and filter listings (66.7%).
Q9: Mobile or desktop access?	100% want mobile access. 57.1% say it is essential; 42.9% say it would be nice.
Q10: (juniors only) Want skills data?	All junior respondents said yes. Some asked about CGPA thresholds.

Appendix C – Product Backlog (Top 9 User Stories)

ID	User Story	Priority	Sprint
US-01	"I want to register and log in securely as this website will contain very sensitive data about me"	Must Have	Sprint 1
US-02	"Having a place where i can upload my resume, list my skills etc would be nice"	Must Have	Sprint 1

US-03	“Having our company vetted out by an admin and giving us a verified option would encourage students to apply to us”	Must Have	Sprint 1
US-04	“Searching internships by location, skills required and timeline would be amazing to sort”	Must Have	Sprint 2
US-05	As a company, I want to post a listing with all relevant details and a closing date.	Must Have	Sprint 2
US-06	As a student, I want to apply with one click if my profile is complete.	Must Have	Sprint 3
US-07	As a student, I want to see the status of each application I have submitted.	Should Have	Sprint 3
US-08	As a student, I want to see the most-requested skills from active postings.	Could Have	Sprint 4
US-19	As a student, I want to post questions and read peer advice in a discussion forum.	Could Have	Sprint 4

Appendix D – Figma Prototype

10 Figma prototype Design built by Jobayer.

Link: <https://www.figma.com/design/NXkL58PnzKf3b6JHXL7Sa3/LOOOOOW-FIDELITYYYY?node-id=0-1&t=Lt9p4uHJS2v55whn-1>

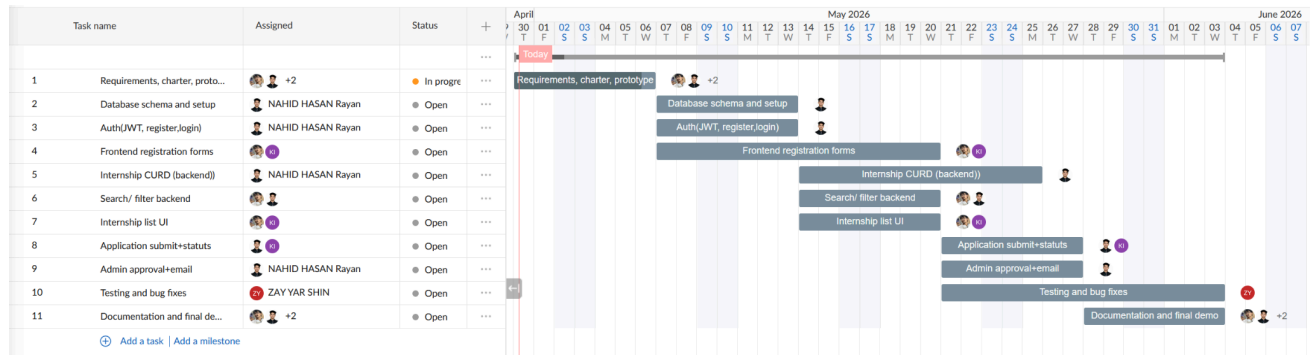
#	Screen	description
1	Browse Listings	search and filter internships by industry, location, and duration
2	Internship Details	full job description, responsibilities, requirements, and Apply button
3	Application status summary	application status summary, recent applications, and interview count
4	Application Form	CV upload, cover letter input, availability date, and submission
5	Supervisor Dashboard	applicant roster, pending reviews, and interview scheduling

Appendix E – Project Schedule: Gantt Chart and PERT

E.1 Gantt Chart

Task	Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Owner
Requirements, charter, prototype	■					All
Database schema and setup		■				Nahid
Auth (JWT, register, login)		■				Nahid
Frontend registration forms		■	■			Khalid, Jobayer
Internship CRUD (backend)			■	■		Nahid
Search / filter backend			■			Nahid, Jobayer
Internship list UI			■			Khalid, Jobayer

Application submit + status						Nahid, Khalid
Admin approval + email						Nahid
Testing and bug fixes						Ghazi, Shin, Nahid
Documentation and final demo						Shin, Khalid, All



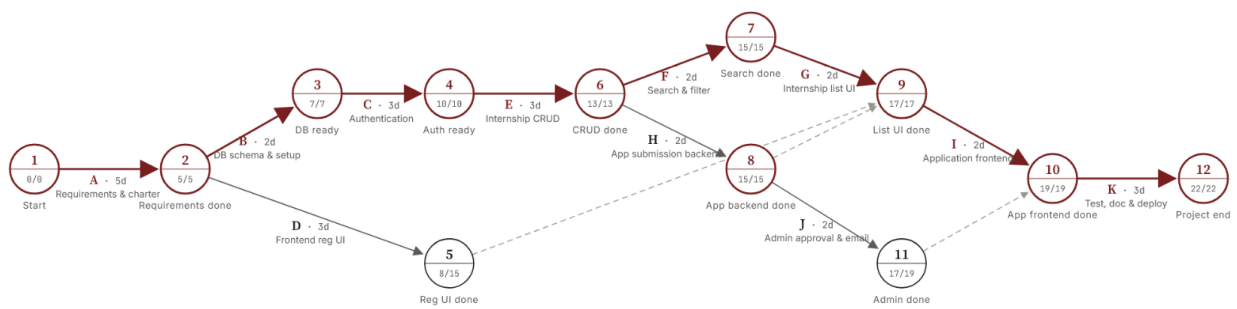
Pictured: Gantt Chart Made with GanttPro

LINK to view the chart in GanttPro

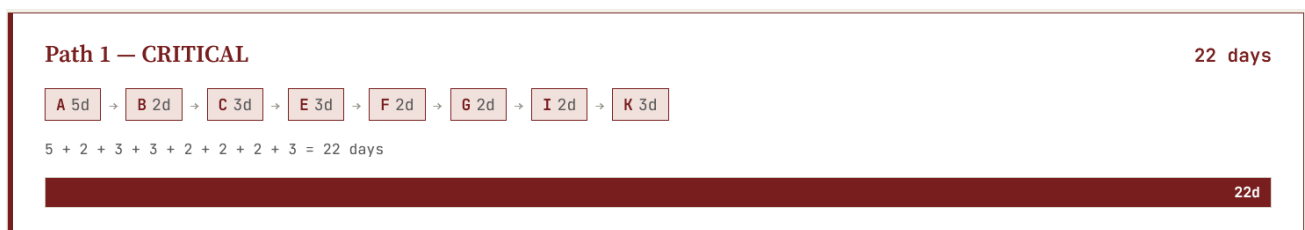
<https://app.ganttpro.com/shared/gantt/c294614f748271872f5b7061b86d50f366461de78c4be84e73384c5603baabc1/2833997>

E.2 PERT Activity Table

ID	Activity	Predecessor	Duration	Assigned To
A	Requirements, charter, and prototype	None	5 days	All
B	Database schema and setup	A	2 days	Nahid
C	Authentication (JWT, register, login)	B	3 days	Nahid
D	Frontend registration UI	A	3 days	Khalid, Jobayer
E	Internship CRUD backend	C	3 days	Nahid, Jobayer
F	Search and filter backend	E	2 days	Nahid, Khalid
G	Internship list UI	D, F	2 days	Khalid, Jobayer
H	Application submission backend	E	2 days	Nahid
I	Application frontend	G, H	2 days	Jobayer
J	Admin approval and email notifications	H	2 days	Nahid
K	Testing, documentation, deployment	I, J	3 days	Ghazi, Shin, Nahid



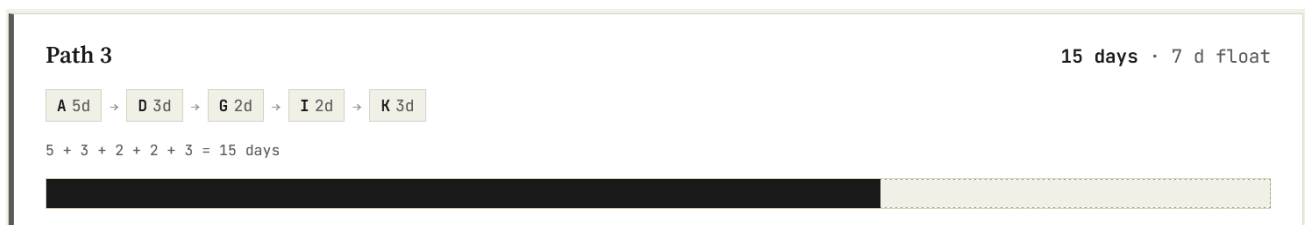
Path 1 (Critical): A -> B -> C -> E -> F -> G -> I -> K (5+2+3+3+2+2+2+3 = 22 days)



Path 2: A -> B -> C -> E -> H -> J -> K — 5+2+3+3+2+2+3 = 20 days (2 days slack/Float)



Path 3: A -> D -> G -> I -> K — 5+3+2+2+3 = 15 days (7 days slack/Float)



Critical path (22 days) runs through all Nahid-owned backend tasks. Frontend work has 7 days of slack.

Appendix F – Database Schema (SQL Reference)

Full CREATE TABLE statements with indexes and foreign keys are in the GitHub repository under <https://github.com/Baxia420/VIP>

Table	Key Fields	Notes
users	id, email, hashed_password, role, name, created_at	Single login table for all roles
student_profiles	user_id (FK), matric_no, cgpa, resume_url, skills	One-to-one with users; skills as text array
company_profiles	user_id (FK), company_name, description, verified_by_admin	Cannot post until verified_by_admin = true
internships	id, company_id (FK), title, description, required_skills, duration, deadline, status	Status: draft active closed
applications	id, student_id (FK), internship_id (FK), status, applied_at, updated_at	Status: pending shortlisted rejected hired

Appendix G – Risk Register

Risk	Prob.	Impact	Mitigation	Owner
Backend delay on auth / DB	Medium	High	Nahid owns and starts first; Firebase Auth as JWT contingency	Nahid
Scope creep	High	Medium	PO approval + sprint capacity check before any backlog addition	Nahid (PO)
Uneven workload	Medium	Medium	Daily standups; SM redistributes blocked tasks	Jobayer (SM)
Limited company input	High	Low	Two confirmed contacts; simulated data as fallback	Dev Team
Deployment issues near deadline	Low	High	Deploy in Sprint 3 so problems surface early	Ghazi, Shin
Docs falling behind	Medium	Medium	Written each sprint; Shin + Khalid co-own	Shin, Khalid

Approval and Acknowledgement

All team members have reviewed and agreed to the roles, responsibilities, and recommended technical approach described in this document.

Nahid Hasan Rayan	Jobayer Alam	Ghazi Anwar Bin Ahmad Razif	Khalid Waleed Issa	Zay Yar Shin
A25MJ0186 Product Owner	A25MJ0184 Scrum Master	A25MJ5072 Dev Team	A25MJ0185 Dev Team	A25MJ0187 Dev Team

Date of Submission: 30 April 2026

Lecturer / Evaluator: Dr. Siti Nur Khadijah Aishah Binti Ibrahim